ESSENTIAL LECTURE 02

Working with Text, Images and Colour

Make graphics clear, text readable and colour purposeful.

What you should be able to do

These are the essential skills for this classroom session.

01 Recognise image types

Explain why photographs and logos may need different graphic formats.

02 Make text readable

Choose clear type, spacing and a sensible heading-to-body hierarchy.

03 Use colour deliberately

Choose a small palette with readable contrast and appropriate emphasis.

04 Improve a notice

Identify weaknesses and sketch a clearer version for readers.

What makes a publication easy to read?

Graphic selection

A photograph explains what happened or what a place looks like. A logo identifies an organisation. Each graphic should support the message.

Text arrangement

Headings, body text and labels should make information easy to scan. Size, spacing and alignment tell readers where to look first.

Colour works with both: it can attract attention, organise sections and reinforce meaning.

Three decisions before you design

IMAGE

Which picture, drawing or symbol helps the reader understand the message?

TEXT

What is the headline? Which details must remain easy to find?

COLOUR

Which colours improve contrast and support the intended mood?

A good design makes these three decisions work together—not compete for attention.

What is a raster image?

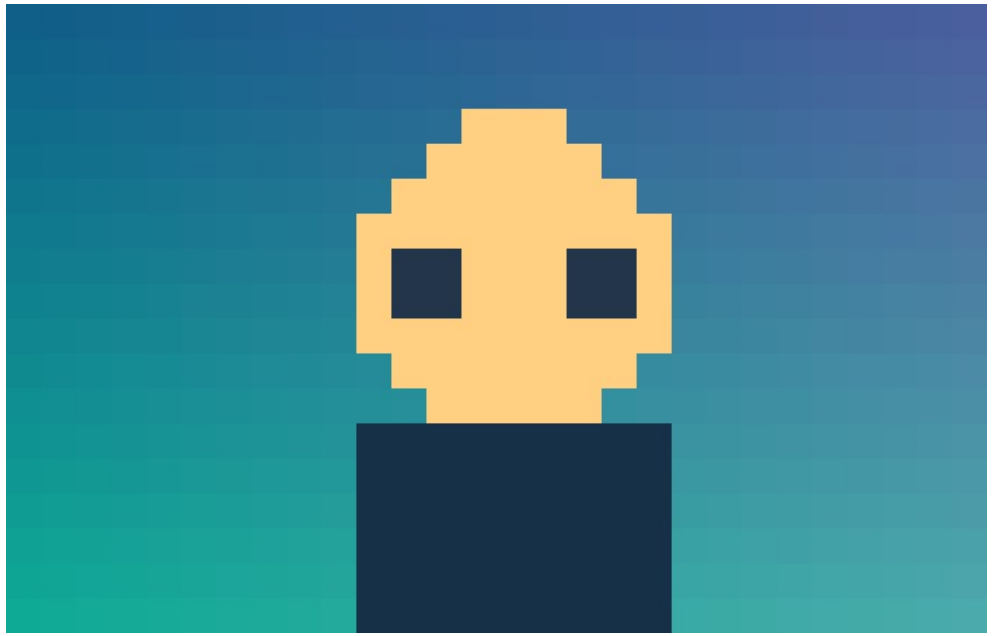
A raster image is made of many small picture elements called pixels. Each pixel stores colour information; together the pixels form a photograph or digital picture.

Examples: phone photographs, scanned pages and most screenshots.

Raster images are particularly useful for detailed textures, shadows and natural colour changes.

Pixels form a picture

A raster image looks continuous at normal size, but it is built from coloured squares.



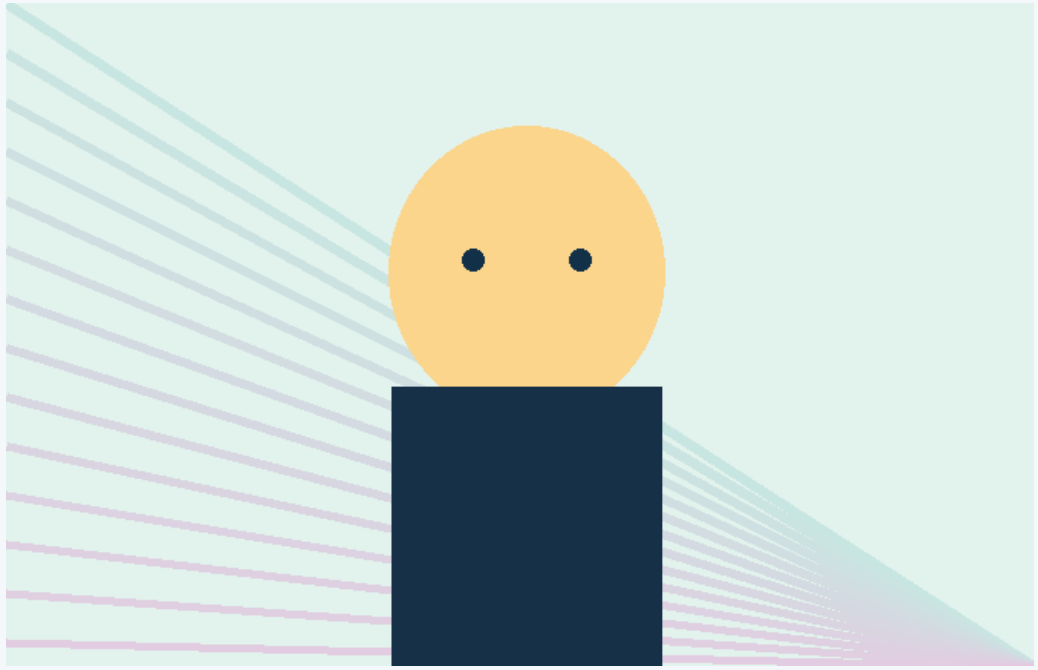
When you enlarge too far

Individual pixels can become visible. Software may smooth the blocks, but smoothing cannot recover detail that was never captured.

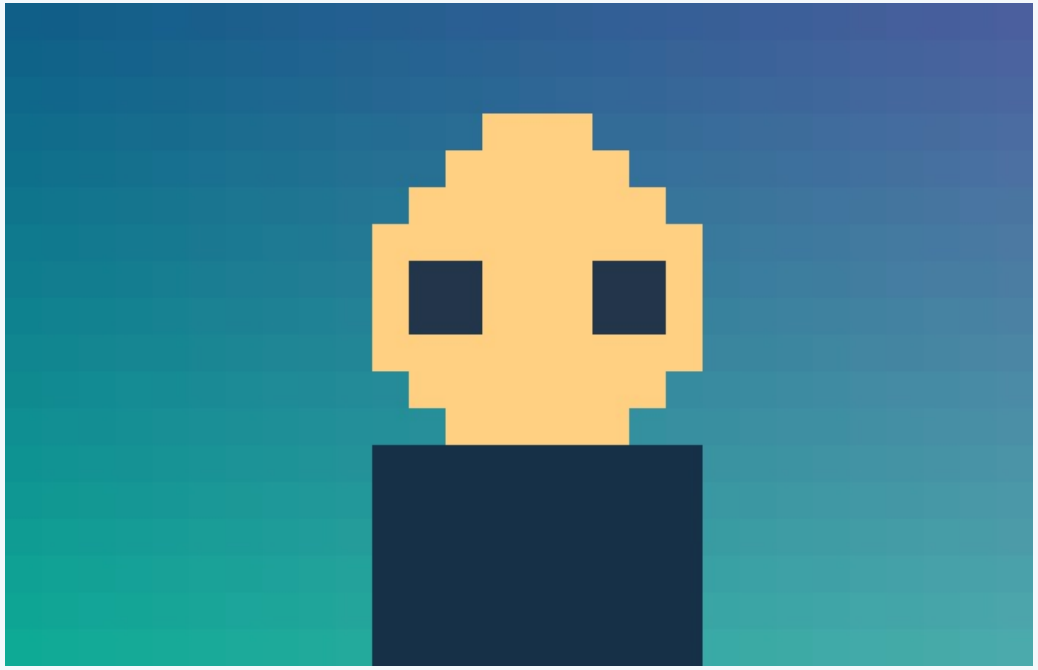
This is called pixelation.

Why image size matters

More original detail



Enlarged low-resolution version



A large printed photograph needs enough pixels for its intended physical size.

What is a vector graphic?

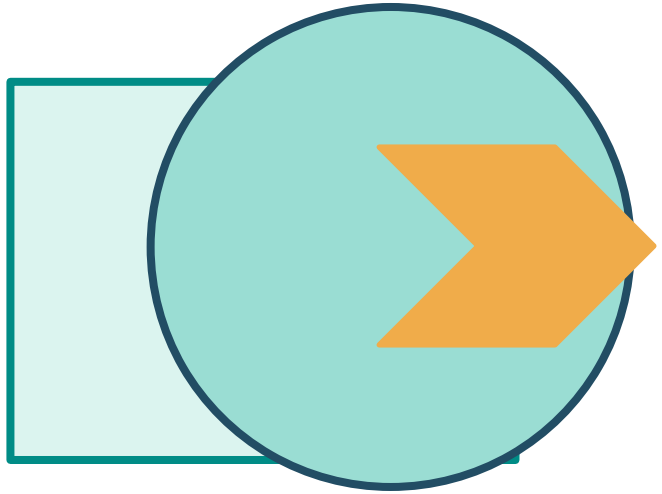
A vector graphic is made from points, lines, curves and shapes described mathematically. These shapes can be scaled without the blocky edges associated with enlarged raster images.

Examples: logos, simple icons, diagrams and illustrations.

A vector file may also contain an embedded raster photograph, so check its actual content.

Vectors use paths and shapes

Changing a vector shape means changing its geometry, rather than colouring individual pixels.



Editable objects

Why scaling is different

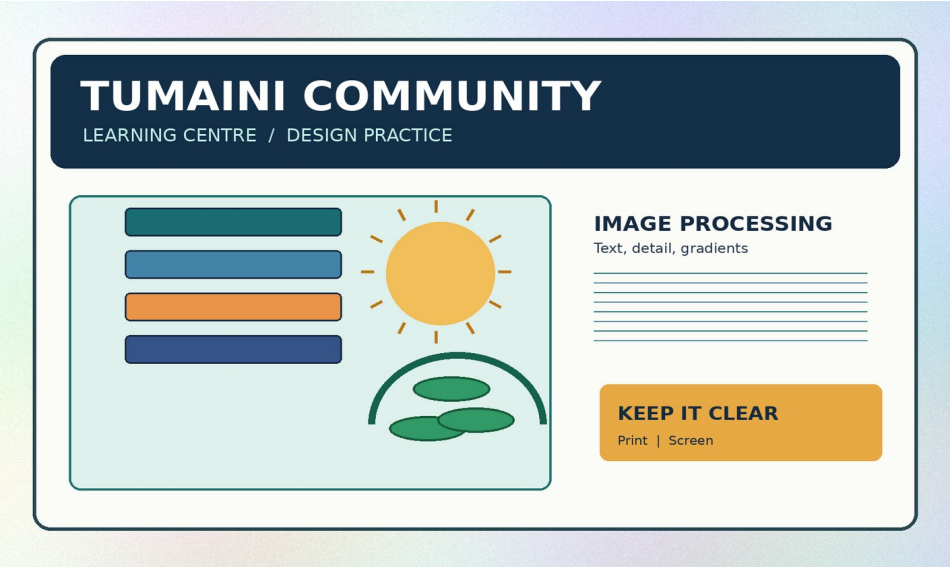
The software redraws each path at the new size. The icon can be displayed on a small badge or a large sign while retaining smooth edges.

Raster versus vector at a glance

	RASTER	VECTOR
Best for	Photographs and detailed images	Icons, logos and simple artwork
Built from	Pixels	Paths, points and shapes
Enlargement	Can show pixelation	Usually stays crisp
Typical formats	JPEG, PNG	SVG, AI

Choose the right graphic for the job

The same publication often includes both raster and vector content.



Workshop photograph or illustration

TUMAINI
LEARNING

Organisation identity

Use detailed raster images for scenes; retain editable vectors for simple symbols and logos.

Four file formats you will meet often

JPEG / JPG

Usually suited to photographs. Small files are possible, but saving with lossy compression can discard image detail.

PNG

Useful for sharp graphics and transparency. PNG uses lossless compression for its stored pixels.

SVG

An editable, scalable vector format suited to simple drawings and logos. It can contain embedded raster images.

A PDF is a document container: it may hold text, raster images and vector shapes.

JPEG and PNG are not interchangeable

Choose JPEG when …

You are sharing a photograph that has many colours and gradual transitions. Adjust the quality setting and check whether the result still looks acceptable.

Choose PNG when …

You need transparent backgrounds, clear edges or text-like artwork. A PNG is still raster-based even when its edges look sharp.

Do not repeatedly save a working photograph as a low-quality JPEG; keep an original or editable source.

What about SVG and PDF?

SVG: an artwork format

SVG stores shapes and paths. Use it for a scalable icon or logo when the receiving software supports it. Keep the original design editable.

PDF: a publication format

PDF is commonly used for sharing completed layouts and print proofs. A PDF can contain several kinds of content; the extension does not guarantee high quality.

Always inspect the final exported file: filenames do not prove that the content is correct.

Quick format decisions

A school-event photograph	JPEG is a sensible starting choice.
A logo with no background	SVG, or a PNG when a raster file is required.
A finished two-page hand-out	PDF for reliable sharing and printing.

Pixel dimensions are not file size

Pixel dimensions

A 1,200 × 800 pixel image contains 960,000 pixels. These values describe its image dimensions, not the size of the saved file on a drive.

File size

File size depends on dimensions, format, compression settings and other data. Two 1,200 × 800 pictures can occupy very different amounts of storage.

A smaller file is useful only if it still meets the publication’s quality needs.

Cropping and resizing are different

Cropping changes what is visible

Cropping removes content near the edges to improve composition or fit a different shape. An unneeded background can disappear.

Resizing changes dimensions

Resizing changes the pixel dimensions or intended physical size. It does not necessarily remove content, but enlarging may reduce apparent quality.

Before editing an important image, save a working copy and preserve the original.

What is typography?

Typography is the selection and arrangement of written characters so that information is readable, organised and appropriate for its audience.

Typography includes typeface choice, size, weight, alignment, spacing and line length.

Good typography is not only decorative: it helps the reader find and understand information.

Typeface, font and style

Typeface

A typeface is a family of letter designs, such as Arial or Georgia. A serif design has small finishing strokes; a sans-serif design does not.

Font and style

A font is a particular implementation or variation of a typeface, such as a bold or italic style. In everyday software, the words are often used loosely.

For beginner publications, one readable type family with clear sizes and weights is often enough.

Serif and sans-serif: compare

Aa Gg Rr

SERIF TYPE

Example: Georgia. The letter ends have small finishing strokes.

Aa Gg Rr

SANS-SERIF TYPE

Example: Arial. The letterforms have simpler endings.

Neither family is automatically better. Choose a typeface that suits the reader and the setting.

Create a clear text hierarchy

Readers should be able to find the most important detail first.

DIGITAL SKILLS

Open training day

Learn to use a computer, write documents and organise files.

Saturday • 10:00 • Main Hall

How hierarchy works

The title attracts attention. The second line explains the event. The paragraph gives context. The smaller line provides details.

Make body text comfortable to read

SIZE

Use sufficiently large body text for the intended medium. Small text may be difficult to read on a classroom screen or printed notice.

SPACING

Leave clear space between lines and separate different kinds of information. Crowded text slows down reading.

ALIGNMENT

Keep related text aligned consistently. For ordinary paragraphs, left alignment is often easier to follow than centring every line.

Readability depends on size, spacing, contrast, line length and the viewing conditions.

A few consistent styles beat many fonts

Too many styles

Several unrelated typefaces, colours and sizes create competition. Readers spend effort decoding the page rather than understanding the message.

A restrained system

Use one type family, perhaps two when necessary. Give headings a clear size or weight difference, then keep body text consistent.

Bold text should identify important information—not every sentence.

Typography can guide or confuse



UNCLEAR



CLEARER

Why the improved notice reads better

HEADLINE

The event name is visibly larger than supporting text. Readers recognise the main message first.

GROUPING

Date and time appear together. Information with the same purpose occupies the same area.

CONSISTENCY

The notice uses a limited set of type styles, clear alignment and enough empty space.

Typography and layout principles from Lecture 1 work together here.

What does colour do in DTP?

Colour is a visual tool that can attract attention, separate sections, improve recognition and create a particular mood. Its purpose should support the message.

Example: one accent colour can draw attention to the date of an event while darker text keeps the details readable.

A colourful page is not necessarily a clear page; readability comes first.

Three useful colour properties

HUE

The colour family we recognise, such as red, yellow, green or blue.

SATURATION

How vivid or muted a colour appears. Highly saturated colours stand out but can be tiring if overused.

LIGHTNESS

How light or dark a colour appears. Differences in lightness often help make text easier to read.

You do not need advanced colour mathematics to choose a practical classroom palette.

RGB and CMYK describe different uses

RGB — light on screens

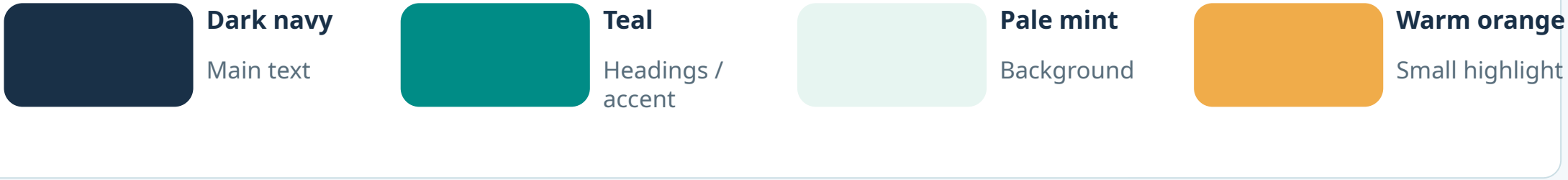
Red, green and blue light are combined by digital displays. RGB is a common colour mode for photographs and digital graphics.

CMYK — printing inks

Cyan, magenta, yellow and black inks are used in many printing processes. Colours may shift between screen and paper depending on devices and settings.

For beginner work, choose suitable export settings and inspect proofs instead of memorising colour profiles.

Build a small, usable palette



A reliable starting rule

Choose one dark colour for most text, one light background, and one or two accents. Repeat those choices consistently rather than introducing a new colour for every line.

Check contrast before decoration

LOW CONTRAST

Important information should not disappear into the background.

HIGH CONTRAST

Dark and light colours can create a clear reading difference.

Colour contrast must be checked for body text as well as headlines.

Do not communicate with colour alone

Weak instruction

“Students in red must submit today.” A reader may not distinguish the colours, or may be viewing a black-and-white printout.

More accessible instruction

“Submit today” with a visible label or icon and adequate contrast. Colour may reinforce the priority, but the words carry the message.

Also check whether your notice still makes sense when photocopied in greyscale.

Bring text, graphics and colour together

IMAGE

Use one relevant picture or symbol. Choose a suitable file format and avoid stretching a small image far beyond its size.

TEXT

Use a strong heading and a short supporting explanation. Group date, location and action details logically.

COLOUR

Use restrained colours, consistent highlights and clear text/background contrast.

The reader should recognise the purpose and locate the key information in a few seconds.

Design review: what is going wrong?



Ask these questions

1. Can you find the main message quickly?

2. Are the date and time easy to locate?

3. Which colours or type styles make reading harder?

4. What should be grouped together?

A clearer version of the same notice



Notice the changes

A clear headline starts the reading order. The message is shorter. Date and time sit together. The limited palette gives the notice a consistent appearance.

Your practical: redesign a notice

01	**Read**	Identify the audience, message and action.
02	**Plan**	Sketch a heading, supporting line and details.
03	**Choose**	Select one or two type styles and a limited palette.
04	**Explain**	Give three reasons why your version is clearer.

Self-check before submitting your sketch

Content

Can a reader identify the title, date, time and organisation without searching? Is any essential information missing?

Design

Is the heading clearly larger than the body? Are alignments consistent? Is contrast sufficient? Have you avoided too many fonts or colours?

You may complete this first design on paper; software is not required for the essential activity.

Essential terms to remember

Raster	Pixels; photographs
Vector	Paths; scalable icons
Typography	Readable arrangement of text
Contrast	Difference that improves visibility
JPEG	Common lossy photo format
PNG	Lossless raster with transparency
SVG	Scalable vector graphic
Hierarchy	Order of importance on a page

Check your understanding

1. Explain how raster and vector graphics differ.
2. Why is JPEG often suitable for photographs, but SVG useful for logos?
3. Name three typography decisions that improve readability.
4. Give two ways colour can help a reader understand a notice.
5. Why should colour not be the only way to communicate urgency?

Next: planning a publication

A good publication begins with a plan.

In Lecture 3, you will turn a brief into a short content outline and a simple page sketch. Bring your redesigned notice: it is an early example of how layout decisions follow the needs of a reader.